Ultra-Lean Plan for a Perplexity-Style Research Tool (100% Open-Source, < 2 weeks build-time, ≈ US $0 infra if run on a single free-tier VM)

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1. Core Idea
   "Glue together" three recent breakthroughs that remove the heavy lifting:

- **Kimi-K2-Instruct 32 B** – open-weight, agent-ready reader that rivals GPT-4 but can run quantised on CPU.
- **SearXNG Meta-Search** – self-hosted metasearch proxy that returns Google/Bing/Brave results without API fees.
- **Firecrawl v2** – OSS crawler that fetches pages in one call and outputs cleaned Markdown + OpenAI-style embeddings.

These three pieces already perform search, clean-up, embedding, and answer generation; you only have to orchestrate.

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2. Minimal Architecture (Four Containers)

```
┌──────────┐      q          ┌─────────────┐
│  React   │ ──────────────▶ │  FastAPI    │
│  UI      │                 │  Orchestr.  │
└──────────┘◀──────────────  └─────────────┘
                                  │
         answer + cites           │
                                   │1. /search?q=…
                                   ▼
                             ┌──────────────────┐
                             │   SearXNG        │
                             │ (meta-search)    │
                             └──────────────────┘
                                   │2. SERP
                                   ▼
                             ┌──────────────────┐
                             │  Firecrawl v2    │
                             │  (crawl + embed) │
                             └──────────────────┘
                                   │3. chunks+vecs
```

```
                 ▼
        ┌─────────────────┐
        │ sqlite-FAISS DB │
        └─────────────────┘
                 │4. top-k
                 ▼
        ┌─────────────────┐
        │ Kimi-K2-vLLM    │
        └─────────────────┘
```

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3. Build Steps (≤ 14 calendar days)

Day	Task	OSS Component	Effort
1	`docker compose init` four services	–	0.5 d
2	Spin up **SearXNG**; tweak engines list	SearXNG	0.5 d
3	Add `/search` endpoint in FastAPI → hits SearXNG and returns top10 URLs + snippets	FastAPI	0.5 d
4-5	Call **Firecrawl** for each URL, get `clean_text` & built-in `embedding`	Firecrawl v2	1 d
6	Store embeddings + meta in lightweight **sqlite-faiss** table	faiss-cpu	0.5 d
7	Implement `retrieve(query)` → embed with **Firecrawl embed model** (already local) → k-NN in faiss	–	0.5 d
8-9	Run **Kimi-K2** under `vllm`-cpu-quant (4-bit) – prompt = user query + top-k chunks + "cite by title" instruction	vllm	1 d
10	Pipe answer back to UI with markdown cites `(1)(2)…`	React	0.5 d
11	Add caching layer (`diskcache`) for SERPs and Firecrawl fetches	diskcache	0.5 d
12	Rate-limit SearXNG (1 req / sec) & rotate free engines (Brave, Google, DuckDuckGo)	–	0.5 d
13	Build Docker CI + GitHub workflow	GitHub Actions	0.5 d
14	Polish UI, deploy to free-tier Fly.io or Oracle Cloud Arm VM	–	0.5 d

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4. Cost Profile

Resource	Choice	Monthly Cost
VM	Oracle Cloud Free Arm 4 vCPU / 24 GB	**$0**
Storage	sqlite file on VM (up to 20 GB)	$0
Traffic	SearXNG scrapes (≈ 8 GB/mo)	$0 on most free tiers
GPU	Not needed (Kimi-K2 4-bit on CPU)	$0

If you want faster answers, rent one **T4 GPU spot** at RunPod for ≈ US $0.12/hr and scale down at night.

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5. Quality Boosts (Optional, still free)
6. **Switch Embeddings**: Use **bge-base-en v1.5** (Microsoft) – best open embedding as of Aug 2025.
7. **Rerank**: Add **ColBERT-v2** CPU reranker on the top 20 chunks for sharper context.
8. **Streaming Answers**: Enable server-sent events from vLLM → UI for progressive display.
9. **PDFs/YouTube**: Firecrawl now parses PDFs & transcripts; simply toggle flags.
10. **Multilingual**: Swap Kimi prompt to requested language; Firecrawl embeddings are language-agnostic.

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6. Why This Is “Perplexity-Grade”

Feature	Perplexity	Your Tool

Web search	Bing API (paid)	SearXNG meta-search (free)
Long-context reader	GPT-4o	Kimi-K2 (32k ctx)
Citations	Yes	Yes (chunk URL mapping)
Follow-up chat	Yes	Add memory dict in FastAPI (10 loc)
Cost	Proprietary	**$0 infra, $0 API**

The recent release of **Firecrawl v2** (embeds while scraping) and **Kimi-K2** (GPT-4-class reasoning on CPU) removes the last paid dependencies. With just four Docker containers and two weeks of coding, you can replicate Perplexity-style research—entirely free.